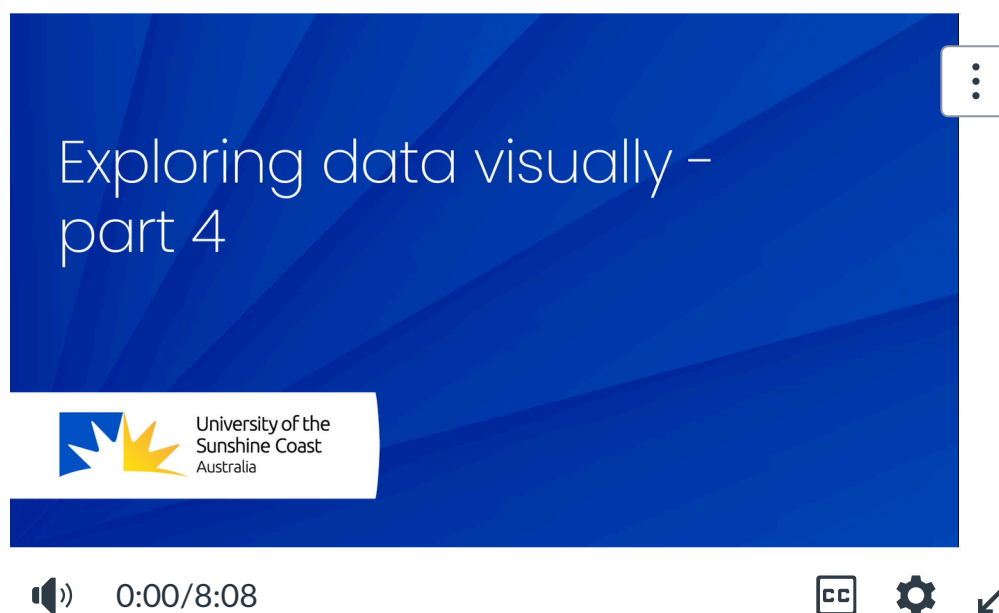


7.6 Visualising geospatial data ↗

□ Watch: Exploring data visually (part 4) [8.08 mins]

Watch this video, which explains what geospatial data is and describes the strengths and weaknesses of various types of map visualisations.

This is useful as it provides understanding of where and how map type visualisations may be used to advantage.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138653?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Exploring data visually \(part 4\)](https://learn.usc.edu.au/courses/35242/files/3168168?wrap=1) <https://learn.usc.edu.au/courses/35242/files/3168168?wrap=1>

Geospatial data is data that includes and represents records about the geographic location of features of the surface of the Earth. Displaying geospatial data on a map can enable the discovery of patterns that might be difficult to interpret in other formats.

□ Watch: Interactive data visualisation for spatial analysis [3.41 mins]

This video is useful because it provides an introduction on how data visualisation techniques can be used to explore, analyse and display spatial data.

As you watch, consider:

- how the ability to explore spatial data may improve the final data visualisations that are created
- why spatial analysis can result in a better audience experience.

This video is private

If the owner of this video has granted you access, please sign in.

Sign in

Source: © ArcGIS (2017).

Choropleth maps

Choropleth maps use colour and/or symbols to depict the values of categorical or quantitative variables that are associated with a geographic area. Their strength is their ability to show the high-level characteristics of a variable in relation to geographic position. They are better for visualising relative comparisons of magnitude than they are at illustrating absolute measures of magnitude. Weather maps are a typical example of a choropleth map.

Choropleth map visualisations are most accurate and useful when the variable of interest is comparatively constant within the geographic areas to be displayed. Where this is not the case, the issue may be mitigated by either:

- variable measures being density based, or
- having the different areas of roughly equal size (making no one area visually distracting).

Another difficulty with choropleth maps is that if the granularity of the location data is not sufficiently fine, the values of the variable of interest within areas and between areas may not be

represented/interpreted correctly. For example, the map may mask variation across an area and/or may suggest abrupt change between areas where, in reality, the change is more gradual.

□ Watch: A guide to choropleth maps [1.40 mins]

This video is useful it explains how choropleth maps work and what they are useful for.

As you watch consider:

- whether using choropleth maps is likely to increase or decrease cognitive load for the audience
- the ways in which choropleth maps could be useful in enhancing dataset story telling.



Source: © The DataViz Cat (2017).

Cartograms

Cartograms are “map-like” diagrams that use geographic positioning. However, the geometry of the different areas (of the whole map) is distorted in order to visualise the proportion of a variable that corresponds to each geographic area. In other words, in a cartogram, each area will be increased or decreased in size relative to the proportion of the variable of interest. Additionally, cartograms can leverage an audience’s prior understanding of the geography of a region. As a result, a more accurate interpretation of the message being conveyed may be achieved and misleading impressions can be avoided.

An issue with cartograms is that the distortion of the areas may be so great that the standard geography becomes unrecognisable and therefore the geographic positioning is meaningless to the audience.

In such cases, an *equal-area cartogram* may be useful. An equal-area cartogram can make a map more readable by making each geographic area equal in size, while retaining a level of reality with geographic positioning within the region. An equal-area cartogram uses shades of colour for the proportional representation of the variable of interest.

□ Watch: Cartograms [5.36 mins]

This video explains why cartograms distort standard map layout visualisations.

As you watch, consider:

- How can cartograms be used effectively within visualisations without misleading the audience?
- What visualisation techniques might assist in reducing the cognitive load of cartograms?



Source: © Patrick Kennelly (2020).